

STRATIFY 2.0

Business Analytics Bootcamp
Analytics Club and Consult Club
UGAC, IIT Bombay

Project: Dark Matter

QuickCart Dark Store Network Design & Cost-to-Serve Challenge

Objective

You're to recommend where QuickCart Pvt. Ltd., a fictional 10-minute quick-commerce company should open new dark stores and which existing ones it should relocate or shut, with the goal of lowering cost-to-serve per order without lowering on-time deliveries (≤ 10 minutes). Build the sharpest network recommendation you can from the data provided.

Background

QuickCart operates 18 dark stores across three Indian cities: Pune, Hyderabad and Jaipur. Order volume has grown roughly 40% year on year, but contribution margin per order remains negative. The board has asked the COO for a credible path to break-even per order without abandoning the 10-minute promise. Diagnostics point to two dominant problems, which this sprint asks you to solve:

1. Coverage gaps: demand from pincodes beyond a store's serviceable radius, causing late deliveries.
2. Cannibalised catchments: stores opened too close together, splitting volume so one runs below the level needed to cover its fixed cost.

Key Tasks

You are expected to:

1. Build the cost-to-serve metric. Join the order log and store master at order level. Construct a per-order cost-to-serve figure using the standard cost assumptions mentioned in the next section. Store fixed cost is allocated by dividing each store's monthly rent and fixed cost by the orders it fulfilled that month.
2. Profile demand and catchments. Map order density by pincode for each city. Compute each store's serviceable radius using the definition in the next section, then flag uncovered demand pockets and overlapping catchments.

3. Design the network. Using the candidate site list, recommend a short list of new store locations and flag any existing stores for relocation or closure. A clear centre of gravity or simple clustering approach is sufficient.
4. Quantify the impact. Estimate how the recommended network changes average cost-to-serve and coverage versus the current baseline.

Standard Assumptions & Definitions

Cost Component	Assumed Value	How to Apply
Rider base payout	Rs 22 per order	Flat on every order
Distance cost	Rs 7 per km	Store to delivery point, one way
Packaging	Rs 8 per order	Flat on every order
Breach Penalty	Rs 30 per breached order	When delivery time > 10 minutes
Store fixed cost	From store master	Monthly rent + fixed cost / orders that month

Serviceable radius: For each store, compute the straight-line distance from the store to every delivery point it served. Bucket those orders into 0.5 km rings. The serviceable radius is the first ring at which more than 20% of orders breach the 10-minute promise. Orders delivered beyond that radius count as outside the catchment for coverage purposes. For straight-line distance, use the haversine formula. At city scale a flat approximation is also acceptable: 1° of latitude ≈ 111 km, and 1° of longitude ≈ 111 × cos(latitude) km.

Deliverables/Outputs

1. Analysis file (notebook or spreadsheet): Cleaned order-level data with your cost-to-serve column, demand density and catchment maps for the cities in scope, exported as static images. Label columns clearly.
2. Final Report: Must contain a recommendation memo, made up of your assumptions log; recommended new store locations with justification; any relocation or closure calls; and a before vs after table of cost-to-serve and coverage.
3. An 8-10 slide deck as an executive summary presentation for the COO of QuickCart.

Measures of Success

Metric	Target
Cost-to-serve per order	Directionally lower vs. current baseline, with a clearly reasoned estimate
On-time delivery (≤ 10 min)	Held at or above the current baseline in your recommended network
Demand coverage	Majority of order volume falls within a serviceable catchment
Constraint realism	Candidate-site rent and fit-out costs are respected
Clarity & Reasoning	A non-technical reviewer can follow the logic

Data Provided (Simulated)

1. Order log: timestamp, delivery location, fulfilling store, basket value, promised vs. actual delivery time.
2. Store master: store ID, location, monthly rent and fixed cost, go-live date.
3. Candidate site list: roughly 15 shortlisted properties with location, carpet area, monthly rent and fit-out cost.
4. Pincode-level population density: a light geospatial layer.

Preparatory Knowledge Recommended

1. Unit economics / cost-to-serve basics: what goes into the cost of fulfilling one order.
2. Basic plotting and clustering: a scatter plot of lat-long coloured by order density is entirely sufficient.
3. Facility-location intuition: centre-of-gravity as a starting heuristic is enough.
4. Excel (pivot tables, basic Solver) or Python (pandas): either toolchain is fine. No specialised optimisation library is required for the submission.